Problem 3

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Problem 1 *Let $f(x) = |5 - x|$.*

(a) *For $a < 5$, find $f'(a)$.*

Explanation. Recall that

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

When $a < 5$, $0 < 5 - a$ and so $f(a) = |5 - a| = 5 - a$. Since we are considering the limit as x approaches a, (and therefore interested in values of x really close to a), we may also assume that $x < 5$ and therefore $f(x) = 5 - x$. Thus

$$\begin{aligned} f'(a) &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \to a} \frac{(5 - x) - (5 - a)}{x - a} \text{ which has form } \frac{0}{0} \\ &= \lim_{x \to a} \frac{-(x - a)}{x - a} \\ &= \lim_{x \to a} -1 = -1. \end{aligned}$$

(b) *For $a > 5$, find $f'(a)$.*

Explanation. When $a > 5$, $0 > 5 - a$ and so $f(a) = |5 - a| = -(5 - a) = a - 5$. Since we are considering the limit as x approaches a, we may also assume that $x > 5$ and therefore $f(x) = x - 5$. Thus

$$\begin{aligned} f'(a) &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \to a} \frac{(x - 5) - (a - 5)}{x - a} \text{ which has form } \frac{0}{0} \\ &= \lim_{x \to a} \frac{x - a}{x - a} \\ &= \lim_{x \to a} 1 = 1. \end{aligned}$$

(c) *Determine whether $f'(5)$ exists.*

Explanation. If $f'(5)$ exists then $\lim_{x\to 5} \dfrac{f(x)-f(5)}{x-5}$ exists. But when x is close to 5 it may be that $x > 5$ or $x < 5$. So in order to find this limit, we have to check whether the two one-sided limits are equal.
This means, $\lim_{x\to 5^-} \dfrac{f(x)-f(5)}{x-5} = \lim_{x\to 5^+} \dfrac{f(x)-f(5)}{x-5}$
Each of these has form $\dfrac{0}{0}$.

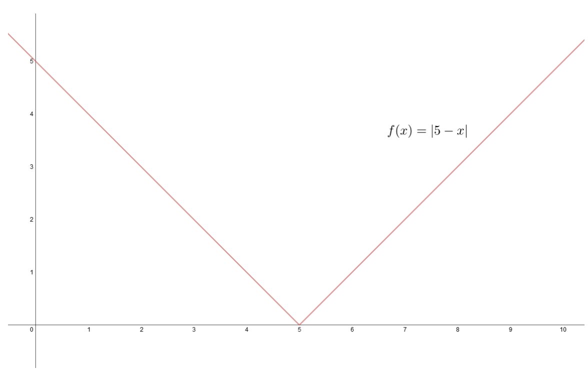
$$\begin{aligned}
\lim_{x\to 5^-} \frac{f(x)-f(5)}{x-5} &= \lim_{x\to 5^-} \frac{-(x-5)-(0)}{x-5} = -1 \\
\lim_{x\to 5^+} \frac{f(x)-f(5)}{x-5} &= \lim_{x\to 5^+} \frac{(x-5)-(0)}{x-5} = 1 \\
\lim_{x\to 5^-} \frac{f(x)-f(5)}{x-5} &\neq \lim_{x\to 5^+} \frac{f(x)-f(5)}{x-5}
\end{aligned}$$

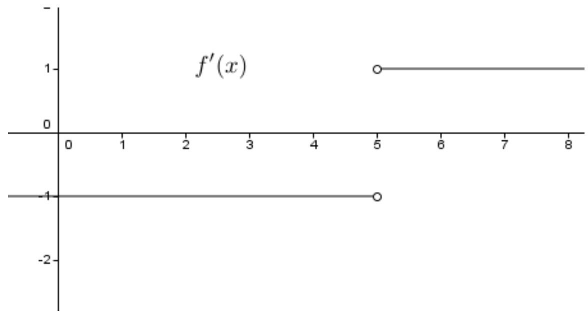
Therefore, $f'(5)$ does not exist.

$$f'(a) = \begin{cases} 1 & \text{if } a > 5 \\ \text{undefined} & \text{if } a = 5 \\ -1 & \text{if } a < 5 \end{cases}$$

(d) *Sketch a graph of the function $f(x)$ and its derivative $f'(x)$*

Explanation.




$f'(x)$
1
0
-1
-2
0
1
2
3
4
5
6
7
8